Great Wall: Coercion-Resistant Authentication

You cannot be forced to hand over something you cannot describe.

Walk-through demo: youtube.com/watch?v=Ry9wzHq1w0A

The Problem: Wrench Attacks Are Rising, and Turning Deadly

Lopp registry, 1 Nov 2026

The public registry of physical attacks on Bitcoin holders lists **801** cases, with a record **170** in 2025 and 39 in Brazil. **About 1 in 25 ends with someone dead** (32 of 801), and **41% of 2025's attacks involved abduction or captivity**, up from 18% in 2021. Press-sourced, so a floor, not a census.

The community's answer has been *"don't tell anyone."* Obscurity is not a mechanism: it is a bet that the attacker does not know something you do, and Kerckhoffs settled that in 1883. When obscurity is the default (when defenses depend on lying, or on the attacker not knowing how they work), attackers eventually learn that escalating violence pays. Where an anti-obscure, i.e. transparent, system leaves the stash operationally out of reach, attackers learn that the attack is futile. Our two papers, each citable by DOI, formalize this:

- **The Deadly Race** (Zenodo, DOI 10.5281/zenodo.22018891): how some self-custody protocols inadvertently give wrench attackers an incentive to kill their victims as means to eliminate a racer to the stash.
- **The Denial Spiral** (Zenodo, DOI 10.5281/zenodo.22778480): why defenses built on denial escalate the attack instead of ending it.

The Solution: Recognition, Not Recall, Behind a Delay

Our protocol is **TGPO, the Time-Gated Perceptual Oracle**, and the property it delivers is **TKBA (Tacit Knowledge-Based Authentication)**. You authenticate by *recognizing* something you cannot describe, so there is nothing to write down and nothing to hand over; a mandated delay means a captive cannot be turned into an immediate transfer. Security is *anti-obscurity*: it does not depend on the attacker being ignorant of it. **One protocol, two deployments**, differing only in who counts failed attempts:

- **Custodial (2^{16}):** the bank counts guesses and freezes the account, as it already does for PINs. Four palettes of sixteen **faces**, reshuffled each use: 65,536 combinations against 10,000 for a 4-digit PIN, and easier to remember. The custodian synthesizes everything server-side.
- **Self-custodial (2^{160}):** nobody counts guesses against your own encrypted seed, so the secret must be high-entropy, derived on your own device from fractal maps, behind a time-lock puzzle.

Anti-obscurity also flips the culture of fear into **"Loud, Proud, and Free"**: an educated attacker knows the attack cannot succeed, so our cards and apparel work like an alarm company's yard sign.

The Business: Three Revenue Streams

Full business model (PDF)

1. **Licensing to banks (B2B, custodial).** Banks and fintechs license the software and run everything on their side; license, seat and integration fees. Serviceable market: **$**80.8**M** across 673 institutions in Latin America. Brazil's central bank already caps night-time Pix transfers: the regulator has conceded that **delay** is the countermeasure. The largest stream, with a 12–18 month sales cycle.
2. **Self-custody (B2C).** The user owns everything and pays for convenience. **Flying Turtle** lets users pay *anonymously* to have their time-lock puzzle solved by Providers on idle compute; we take a platform fee. Serviceable market: **$**19.4**M**. Smaller, and it is where the credibility comes from.
3. **Education (B2C).** Paid live cohorts on coercion-resistant self-custody, launched from a free master class at **Anarcópolis** (16 Nov 2026). Small revenue; it proves people pay and feeds the other two streams.

Status, Honestly

- **Self-custodial wallet** (github.com/Yuri-SVB/great-wallet): public beta. Not yet safe for real funds.
- **Flying Turtle:** in development, release expected within about a month.
- **Custodial deployment:** designed, not built — and the smaller engineering problem.
- **Company:** no entity and no revenue yet; incorporation on term sheet, in the jurisdiction the round prefers.

Why Us

Founder

Yuri S. Villas Boas: author of Bitcoin **BIP-450**; author of The Deadly Race and The Denial Spiral; contributor to the attack registry the model is calibrated on; about to begin a **doctorate on this subject** at PPGCC.

Near-Term Milestones

Master class & pilot course: 16 Nov 2026 • **Flying Turtle release:** Dec 2026 • **First bank pilot:** after the seed

The Ask

Pre-seed, then seed

Now — pre-seed: up to **$**50**K** at a **$**1**M** post-money valuation (e.g. $10K = 1%), convertible into equity on incorporation. Funds the course, the Flying Turtle release and incorporation.

Next — seed: $600**K** for 20% at a **$**3.0**M** post-money valuation: 17 months of runway to build the team, the custodial integration and the first bank pilot.

Demo • App • Business model • The Deadly Race • The Denial Spiral • loudproudandfree.com • yuri-svb.github.io